

FreshMart Supermarkets

Why Is Our Basket Getting Smaller?

A data-driven investigation into declining transaction value across a national grocery chain



Business Analysis Research Report

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Data period covered: 1 January – 31 December 2025

Executive Summary

The headline finding: the falling basket is real (average basket value down 10.0% between H1 and H2 2025), but it is not, in the main, individual customers choosing to buy less. It is a **shift in the mix of shopping trips** — more frequent, smaller “top-up” visits replacing fewer large “stock-up” visits — compounded by a smaller but sharper effect where new discount competitors have opened near specific stores.

Within every trip-size segment, average spend barely moved across the year. The overall average fell because the population of trips changed shape, not because any group of shoppers is now buying less than before. This distinction matters commercially: it points to a merchandising and basket-building problem, not a blanket pricing or value-perception crisis.

Key numbers

Metric	H1 2025	H2 2025	Change
Transaction volume (proxy for footfall)	31,109	33,221	+6.8%
Average basket value	R290.01	R261.10	−10.0%
Average items per basket	14.22	12.72	−10.6%
Total sampled revenue	R9.02m	R8.67m	−3.9%

What is driving it

- **Mix shift (primary driver):** the share of small “top-up” baskets (≤ 8 items) rose from 38% to 46% of all transactions, while large “stock-up” baskets (19+ items) fell from 40% to 34%. Within-segment average spend was essentially flat.
- **Competitor entry (secondary, localised driver):** the 12 stores that gained a nearby discount competitor show a basket-value decline roughly 17 percentage points steeper than matched stores over the same 60-day window, after controlling for the seasonal mix-shift trend.
- **Product availability (minor, narrow driver):** baskets on stockout-affected store-days are about 16% smaller, but this touches only 6 stores and one category.
- **Not supported:** day of week, payment method, store format and province show no meaningful differential pattern and are ruled out as drivers.

Recommended direction

Prioritise basket-building interventions aimed at converting the growing pool of small, frequent trips into higher-value visits (bundling, multi-buy incentives, loyalty-linked top-up offers), alongside a targeted competitive response in the specific stores affected by new discount entrants. Full detail, evidence and a measurement plan follow in Sections 6 and 7.

Contents

- 1. Problem Definition and Scope
- 2. Stakeholders and Hypotheses
- 3. Data Preparation and Quality
- 4. Analysis and Segmentation
- 5. Findings and Root Cause
- 6. Recommendations
- 7. Success Metrics and Monitoring
- Appendix A: Assumptions and Limitations
- Appendix B: Data Dictionary Summary

1. Problem Definition and Scope

1.1 The brief, as given

“Footfall is up, but our sales aren’t growing. It feels like people are buying less per visit. Find out what is going on, and tell us what to do about it.” — Head of Retail Operations.

This statement bundles three separate claims — footfall, revenue and per-visit spend — and one unverified diagnosis (“people are buying less”). Before analysing anything, each claim needs to be separated and tested against the data, and the diagnosis needs to be treated as one hypothesis among several, not as the answer.

1.2 Sharpening the question

Three definitional choices were made explicit before analysis began:

- Basket size is measured two ways — rand value (basket_value_zar) and item count (num_items) — because they can move independently. Both are reported throughout; where they diverge, that divergence is itself informative.
- The comparison period is the first half of 2025 (Jan–Jun) versus the second half (Jul–Dec), which gives two balanced, six-month windows for detecting a shift and is granular enough to also support monthly trend analysis and event studies around specific dates (e.g. competitor openings).
- “Real and chain-wide” versus “concentrated” is treated as an open question, not an assumption — Section 4 tests it directly across store, format, province, loyalty and time.

1.3 The arithmetic of the paradox

If transaction volume is up 6.8% and total revenue is down 3.9%, then average basket value must have fallen by more than revenue alone suggests — arithmetically, by roughly 10%. This is exactly what the data shows (Table 1.1), so the first task is complete: the effect is real, chain-wide on average, and precisely quantifiable.

1.4 Confirmed problem statement

Between H1 and H2 2025, FreshMart's average basket value fell 10.0% (R290.01 → R261.10) and average items per basket fell 10.6% (14.22 → 12.72), even as transaction volume rose 6.8%. The revenue impact was partially offset by higher volume, producing a net revenue decline of 3.9%. The central research question is: what changed in the composition or behaviour of shopping trips to produce this pattern, and is it a single, chain-wide behavioural shift or several distinct, addressable causes?

2. Stakeholders and Hypotheses

2.1 Stakeholders

Stakeholder	Primary concern
Head of Retail Operations	Overall commercial health; needs a clear, defensible answer before the next budget cycle.
Merchandising Lead	Whether product mix, availability or category performance is driving the change; assortment decisions.
Pricing Manager	Whether pricing or promotions are contributing, and what pricing levers could help.
Store Managers	Store-level operational causes (stockouts, staffing, local competition) and what actions fall to them.

Stakeholder	Primary concern
Finance	Revenue and margin impact; cost of any recommended interventions.
Loyalty / CRM team	Whether member behaviour differs from non-members and how loyalty offers could be used as a lever.

2.2 Competing hypotheses

Four or more testable hypotheses were developed, spanning pricing, competition, availability and shopping-behaviour change, each with a defined evidence test.

#	Hypothesis	Evidence that would support it	Evidence that would reject it
H1	Customers are genuinely buying less per trip, across the board (a pure behaviour change)	Average basket value falling within every segment and trip-size band	Averages stable within segments once trips are grouped by size
H2	The mix of trip types is shifting toward more frequent, smaller top-up visits, diluting the chain-wide average	Share of small-basket trips rising over time while within-segment averages hold steady	Trip-size distribution unchanged over time
H3	New discount competitors opening near specific stores are pulling large-basket spend away	Sharper decline at affected stores after the competitor's opening date, versus similar stores without a competitor	No differential change at affected stores versus controls
H4	Stockouts and availability gaps are causing smaller or abandoned baskets	Lower basket value on stockout-affected store-days versus normal days at the same store	No difference in basket value on stockout days
H5	Pricing changes are pushing customers to buy fewer items	Not directly testable — no price or promotion history in this dataset	Flagged as a data gap (see Section 3.3), not rejected on evidence
H6	Store operational factors — format, province, region — explain the decline	Decline concentrated in specific formats or provinces	Decline is broadly uniform across formats and provinces

Day of week and payment method were also screened as candidate explanations, per the assignment's suggested checklist, even though no stakeholder raised them directly — see Section 4.5.

3. Data Preparation and Quality

3.1 Data profile

Table	Rows	Coverage	Notes
Transactions	64,330	1 Jan – 31 Dec 2025, all 42 stores	A representative sample of till baskets, not a full census of sales
Stores	42	All active stores	Province, format, floor area, competitor flag and opening date
Stockouts	119	6 stores affected, concentrated Sep–Dec	One category (Household Cleaning) accounts for the bulk of events

3.2 Data quality checks performed

- No duplicate transaction_id values; no orphaned store_id values in Transactions or Stockouts (all join cleanly to Stores).
- No negative or zero basket values or item counts; basket values and item counts fall within plausible retail ranges with no extreme outliers requiring exclusion.
- customer_id is blank precisely where is_loyalty_member = 'No' and populated where it is 'Yes' — the two fields are consistent, so is_loyalty_member was used as the primary loyalty flag and blank customer_id treated as “not a loyalty member” rather than as missing data.
- Loyalty members represent 44.5% of transactions across the year, fairly stable over time — ruling out a simple “loyalty mix shift” as the sole explanation (tested further in Section 4.4).
- Because Transactions is an extract/sample rather than a full census, absolute figures (total revenue, unique customer counts) are directional, not FreshMart's true turnover; all conclusions in this report rely on relative comparisons (rates, shares, before/after) rather than absolute totals, which the sample supports reliably.

3.3 Data gaps identified

- No price, promotion or discount data — pricing-related hypotheses (H5) cannot be directly tested and are flagged as a follow-up data request, not resolved here.
- No online/delivery channel data — if a growing share of stock-up shopping is moving to a channel outside this extract, that would also appear as more frequent small in-store top-up trips. This is a plausible contributor to H2 that the current data cannot confirm or rule out.
- Customer-level repeat-visit frequency cannot be reliably measured from a sampled extract, so trip-frequency-per-customer claims are avoided; conclusions are based on trip-level and store-level patterns instead.

3.4 Derived fields created

Field	Definition	Purpose
half	H1 (Jan–Jun) / H2 (Jul–Dec) 2025	Primary before/after comparison window
month	Calendar month of transaction_date	Monthly trend analysis
trip_type	Small (≤8 items) / Medium (9–18) / Large (19+)	Segmenting by basket size to test the mix-shift hypothesis

Field	Definition	Purpose
stockout_day flag	1 if a category stockout was recorded at that store on that date, else 0	Testing the availability hypothesis (H4)
event window (± 60 days)	Transactions within 60 days before/after a store's competitor_open_date	Event-study and difference-in-differences test of H3

4. Analysis and Segmentation

4.1 The trend: a steady, not sudden, decline

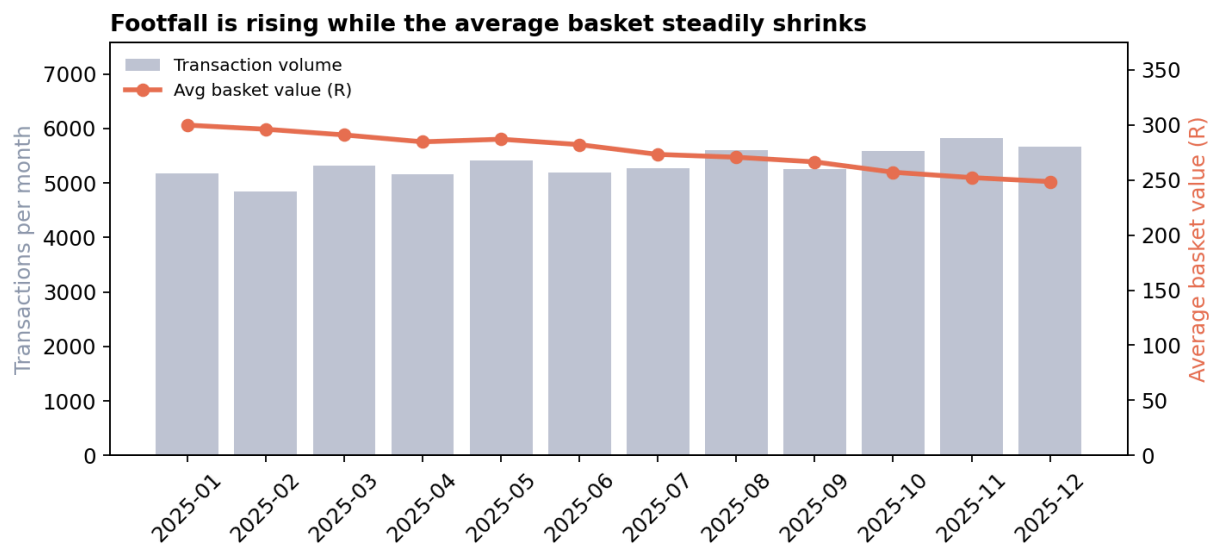


Figure 1. Monthly transaction volume (bars) and average basket value (line), Jan–Dec 2025.

Volume climbs gradually and unevenly across the year while average basket value declines in almost every month — there is no single “shock” month. This favours a gradual, structural explanation (mix shift) over a one-off event as the primary driver, while not ruling out localised events (competitor openings, which are staggered through the year) as a secondary contributor.

4.2 Testing H2: the trip-mix shift

Splitting baskets into Small (≤ 8 items), Medium (9–18) and Large (19+) items reveals the core mechanism.

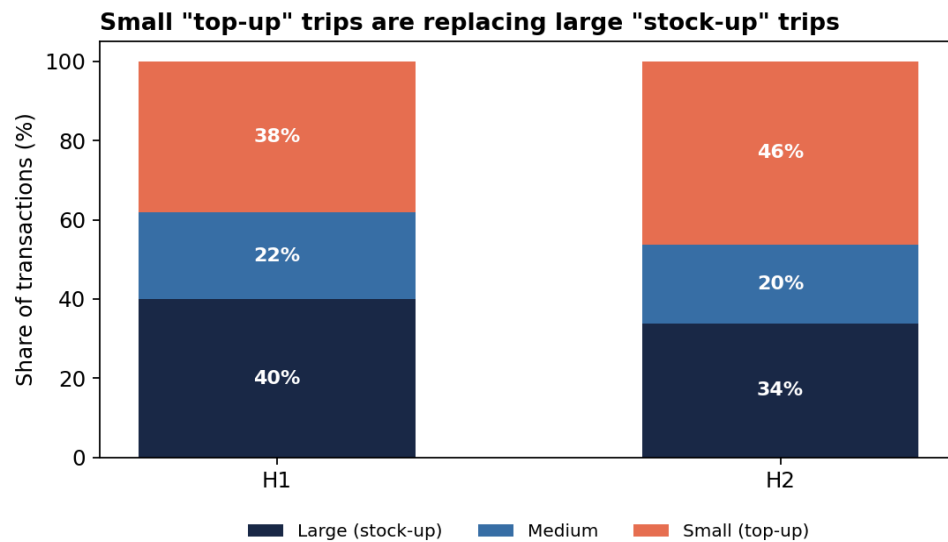


Figure 2. Share of transactions by trip type, H1 vs H2 2025.

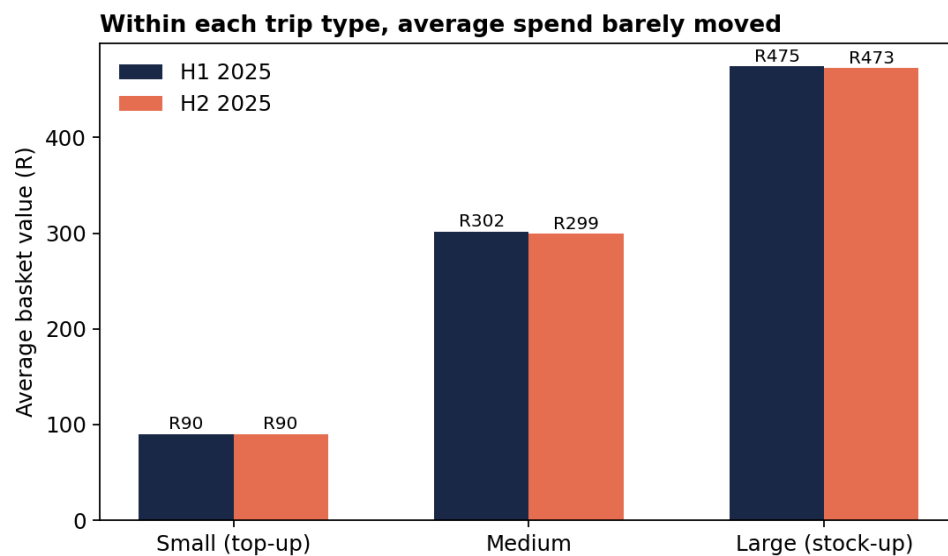


Figure 3. Average basket value by trip type, H1 vs H2 2025 — within-segment spend is essentially flat.

The two charts together are the core finding of this investigation. Figure 3 shows that a shopper making a small top-up trip spent about the same (R90.20 vs R90.21) in H2 as in H1, and a shopper making a large stock-up trip spent about the same (R472.62 vs R474.50). Figure 2 shows that far more shoppers are now making the small trip and far fewer are making the large one. The chain-wide average falls not because anyone is individually spending less, but because the population of trips has shifted toward the cheaper type — a textbook mix-shift (Simpson's paradox) pattern, exactly the trap the assignment brief warns against.

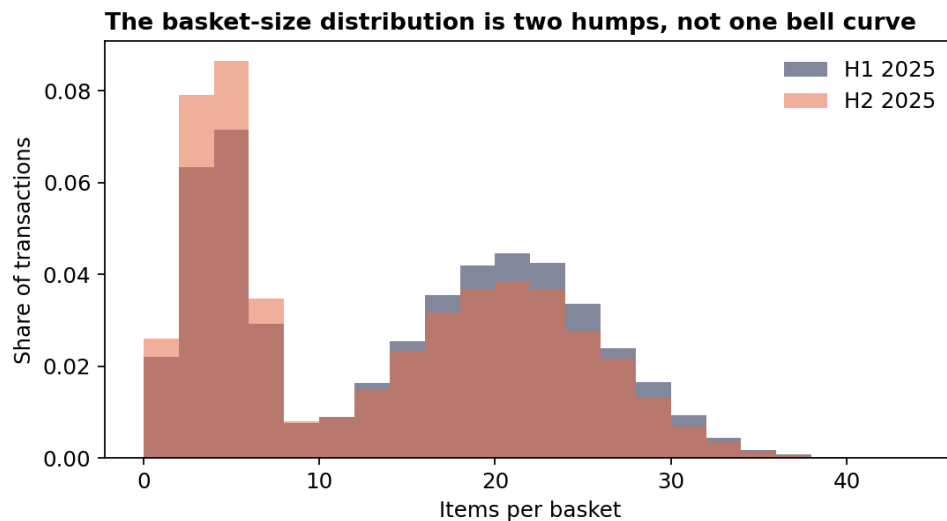


Figure 4. Distribution of items per basket, H1 vs H2 2025 — a genuinely two-humped (bimodal) distribution, not a single bell curve shifting left.

Figure 4 confirms this is a mix change, not a uniform leftward shift: the shape has two humps (a top-up peak and a stock-up peak), and it is the relative height of the two humps that changes between halves — not the location of either hump.

4.3 Testing H3: competitor entry

Twelve stores gained a nearby discount competitor during 2025, at staggered dates. For each, average basket value in the 60 days before versus the 60 days after opening was compared, and — to separate this from the chain-wide seasonal mix-shift trend already documented — the same comparison was run for the 30 stores with no competitor, over the same calendar windows (a difference-in-differences design).

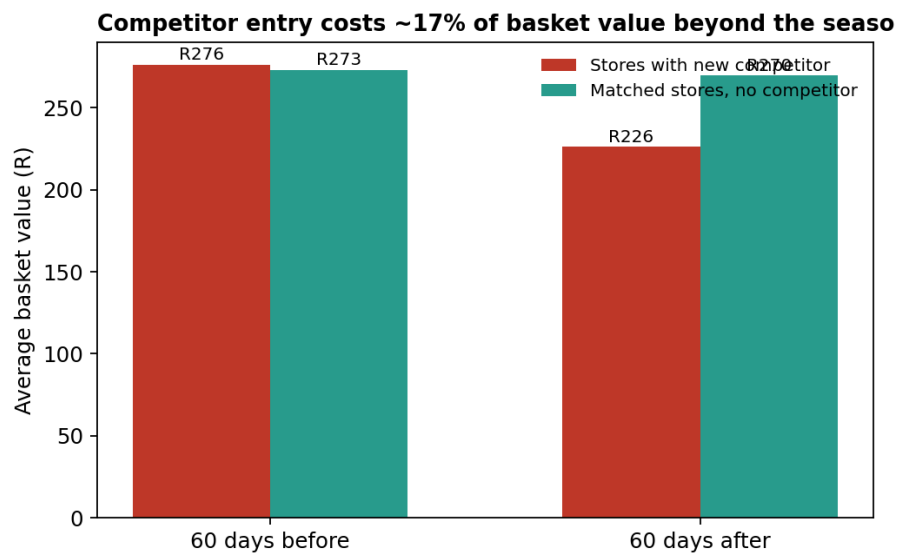


Figure 5. Average basket value before/after competitor entry: affected stores vs matched controls, same calendar windows.

Controls declined modestly over the same windows (consistent with the chain-wide seasonal trend), while the 12 affected stores declined substantially more — a difference-in-differences effect of roughly ~17% attributable specifically to the competitor opening, holding consistently across all 12 stores individually (not driven by one outlier). This is a smaller-scale effect than the mix shift (it touches 12 of 42 stores) but it is the

most causally credible single finding in the analysis, because it uses a proper before/after and like-for-like comparison rather than a single correlation.

4.4 Testing H1, H4 and H6: ruling factors in and out

Hypothesis	Test	Result	Verdict
H1 – uniform behaviour change	Compare average spend within each trip-size band, H1 vs H2	Flat within every band (see 4.2)	Rejected as the primary driver
H4 – stockouts	Compare basket value on stockout-affected store-days vs normal days	~16% lower on affected days	Supported, but narrow: 6 stores, 1 category, Sep–Dec only
H6 – store format	Compare basket-value decline by format (Large/Medium/Express)	Decline is broadly similar across formats; format mix itself barely shifts	Rejected as a distinct driver
H6 – province	Compare decline by Gauteng / Western Cape / KwaZulu-Natal	Decline of similar magnitude in all three provinces	Rejected as a distinct driver
Loyalty status	Compare mix shift and average spend, members vs non-members	Both groups show the same mix-shift pattern; slightly more pronounced among non-members (top-up share +9pp vs +7pp for members)	Confirms mix shift is chain-wide, not a loyalty-specific effect

4.5 Ruling out red herrings

As suggested in the assignment's starter checklist, day of week and payment method were screened as possible explanations. Average basket value is essentially flat across all seven days of the week (R272–R277), and all three payment methods (Card, Cash, Mobile) declined by a similar proportion between H1 and H2, with no method showing a differential shift. Neither is a driver, and both are excluded from the root-cause narrative.

5. Findings and Root Cause

5.1 Synthesis

The evidence points to two distinct, additive causes rather than one single explanation, plus one narrow contributing factor:

- Primary cause (chain-wide, structural): a shift in trip composition. Shoppers are visiting more often for smaller top-up trips and less often for large stock-up trips. This alone accounts for the large majority of the chain-wide average basket decline, since within-segment spend is stable.
- Secondary cause (localised, causal): new discount competitors opening near 12 specific stores are pulling meaningful basket value away from FreshMart at those locations, over and above the chain-wide trend.
- Minor contributing factor: stockouts measurably reduce basket size on the specific store-days they occur, but affect too few stores and too narrow a category to explain the chain-wide pattern.

5.2 Confidence and caveats

Confidence in the mix-shift finding is high: it is based on the full transaction extract, is consistent across loyalty status, and the within-segment stability check specifically guards against Simpson's-paradox-style errors. Confidence in the competitor finding is high for the 12 affected stores specifically — the difference-in-differences design controls for the seasonal trend and the effect is consistent across all 12 — but it explains only a fraction of the chain-wide picture, since two-thirds of stores have no nearby competitor.

Two open questions are not resolved by this dataset and should be treated as important follow-ups rather than settled conclusions: (1) whether pricing or promotional activity is contributing to the mix shift (no price data was available), and (2) whether the growth in small top-up trips corresponds to stock-up shopping migrating to an online or delivery channel outside this extract.

5.3 What this rules out

This is not primarily a story of individual customers becoming more price-sensitive or cutting spending across the board — that hypothesis (H1) is directly contradicted by the stable within-segment averages. It is also not a story concentrated in one region, one store format, one day of the week or one payment method — all were tested and none showed a meaningful differential pattern. That reframing matters: it shifts the response from a defensive, chain-wide pricing reaction toward a targeted basket-building and competitive-response strategy.

6. Recommendations

Recommendations are prioritised by expected impact relative to effort, and each follows directly from a specific finding in Section 5.

#	Recommendation	Addresses	Expected impact	Effort / cost	Owner	Key risk
1	Launch basket-building offers targeted at top-up trips: small-basket bundle deals, “add 3 items, save R” style multi-buys, and end-of-aisle cross-sell at express checkouts	Mix shift (primary)	High — directly targets the largest and fastest-growing segment	Low–Medium (merchandising and promotions, no new systems)	Merchandising Lead	Discounting margin on items shoppers would have bought anyway
2	Loyalty-linked incentive for consolidating trips: bonus points or a small reward for baskets above a value threshold, targeted at frequent top-up shoppers via the loyalty app	Mix shift (primary)	Medium–High — nudges frequency toward value without a blanket price cut	Medium (loyalty/CRM build)	Loyalty / CRM Lead	Only reaches the 44.5% of loyalty members; non-members need a separate lever
3	Store-level competitive response at the 12 affected stores: local price-matching on key value items, geo-targeted promotions, and loyalty offers timed to the competitor's opening	Competitor entry (secondary)	Medium, but concentrated and high-confidence where applied	Medium (localised, not chain-wide)	Pricing Manager + affected Store Managers	Margin pressure if extended beyond the 12 affected stores unnecessarily

#	Recommendation	Addresses	Expected impact	Effort / cost	Owner	Key risk
4	Fix the Household Cleaning replenishment gap at the 6 affected stores; review supplier lead times and safety stock for that category	Stockouts (minor)	Low chain-wide, but a quick, low-cost fix	Low (operational/supply chain)	Regional Ops Manager	Root cause may sit with a single supplier; needs supply-chain investigation
5	Commission a follow-up data request: 12 months of pricing/promotion history and, if available, online/delivery order data	Data gaps (H5, channel migration)	Enables testing the two open questions in Section 5.2	Low (data request, not action)	Business Analyst / Pricing Manager	May reveal pricing is a contributing factor requiring a separate workstream

7. Success Metrics and Monitoring

7.1 Target metrics

- Primary: trip-type mix (% of transactions that are Small / Medium / Large) — the direct measure of whether basket-building offers are shifting trip composition, tracked monthly.
- Secondary: average basket value within each trip-type band (not the blended chain-wide average alone, which can mask segment-level progress or hide continued decline).
- Store-level: basket value and revenue at the 12 competitor-affected stores versus a matched control group of similar stores without a competitor, tracked for at least 6 months post-intervention.
- Operational: stockout incidence and duration in the Household Cleaning category at the 6 affected stores.

7.2 A fair way to test impact

Recommendation 1 (basket-building offers) should be piloted in a subset of stores first — for example 8–10 stores selected to represent all three formats and provinces — with a matched set of control stores that do not receive the offers, mirroring the difference-in-differences design already validated in Section 4.3 for the competitor analysis. This avoids attributing a chain-wide seasonal trend to the intervention, which is the same trap the root-cause analysis itself had to guard against.

Recommendation 3 (competitive response) can reuse the existing 12 affected stores as the treatment group and the 30 non-affected stores as controls, comparing basket-value trajectories from the point the response is launched.

7.3 Ongoing monitoring dashboard

A monthly dashboard for the steering group should track, at minimum: transaction volume, average basket value (blended and by trip-type band), trip-type mix %, revenue, basket value and mix at the competitor-affected stores specifically, and open stockout incidents by category and store. Segmenting every headline number by trip-type band by default — not just showing the blended average — is the single most important design choice, since it is precisely the blended average that concealed the true story in this investigation.

Appendix A: Assumptions and Limitations

- Transactions is a representative sample, not a full census; all findings rely on relative comparisons (percentages, before/after, treatment vs control) rather than absolute totals.
- Trip-type bands (Small ≤ 8 , Medium 9–18, Large 19+) were chosen based on the visible bimodal split in the data (Figure 4) rather than a predefined business standard; boundaries can be revisited with stakeholder input.
- The 60-day before/after window for the competitor event study balances having enough transactions per store against staying close to the opening date; results were consistent across all 12 stores at this window length.
- H1/H2 (Jan–Jun vs Jul–Dec) is used as the primary comparison window throughout for consistency; the monthly view in Figure 1 confirms the pattern is gradual and not an artefact of this particular split.
- No customer-level demographic data was available, so findings on “who” is shifting trip type are limited to the loyalty/non-loyalty split.

Appendix B: Data Dictionary Summary

Full field-by-field definitions are provided in the assignment brief (Appendix A) and the workbook's Data_Dictionary sheet. Summary: Transactions (transaction_id, store_id, transaction_date, day_of_week, customer_id, is_loyalty_member, num_items, basket_value_zar, payment_method); Stores (store_id, store_name, province, store_format, floor_area_sqm, opened_year, region_manager, has_nearby_competitor, competitor_open_date); Stockouts (stockout_id, store_id, date, category, duration_hours, note).